

RTG 3186 / STAUFEN 2026

The cosmic microwave background

I. Expansion, recombination, and sound waves

Daan Meerburg

29 September–1 October 2026

A little about me

Daan Meerburg

CMB cosmology

Current collaborations

- Simons Observatory
- CCAT
- LiteBIRD

Previous collaboration

- Planck

My perspective on the CMB draws on both satellite and ground-based experiments.

Outline for Lecture 1

1. An expanding Universe: distances, energy, and temperature.
2. An opaque plasma becomes transparent.
3. Gravity and photon pressure create sound waves.
4. Recombination captures their phases.



Slides, notes, and
notebook

Goal: understand where the acoustic pattern in CMB comes from.

A 10-minute break near the middle. The QR code opens the public [GitHub repository](#).

From an (un)expected signal to precision cosmology

Late 1940s Alpher and Herman predict a relic radiation bath from a hot early Universe, in the context of Gamow's hot-Big-Bang work.

1965 Penzias and Wilson publish their excess microwave antenna temperature. Dicke and collaborators identify the cosmological interpretation.

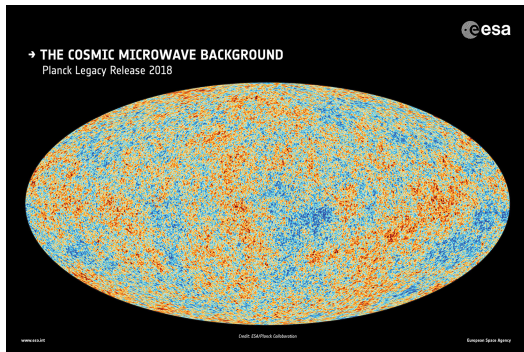
1990 / 1992 COBE establishes the nearly perfect blackbody spectrum (FIRAS), then detects primordial temperature anisotropies (DMR).

2000s / 2010s WMAP and Planck map the sky in increasing detail: the tiny fluctuations become a precision test of cosmology.

Two discoveries: a nearly uniform thermal background, and tiny variations across the sky.

Historical background: [ESA](#); [NASA/COBE history](#); [NASA/LAMBDA](#).

Planck's CMB temperature map



- The whole sky in a Mollweide projection.
- False colour: orange/red is warmer; blue is cooler than the mean.
- Typical differences: $\sim 100 \mu\text{K}$ on a 2.725 K background.

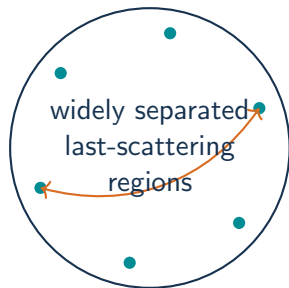
What sizes of hot and cold patches are present? The angular power spectrum makes this quantitative.

© ESA/Planck Collaboration, [Planck Legacy Release 2018](#). Foreground-cleaned CMB reconstruction; mean and motion-induced dipole removed.

The smooth CMB poses an initial-conditions problem

At recombination, ordinary causal processes could coordinate only a limited region.

- Widely separated directions had no time to exchange light in the standard hot-Big-Bang history.
- Yet their mean temperatures agree to about one part in 10^5 .
- Small fluctuations are present over a very broad range of scales.



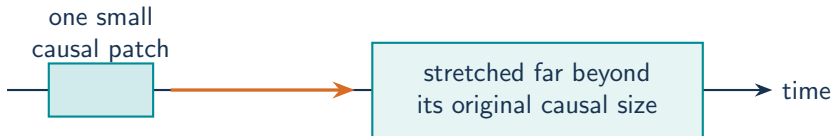
Why did causally disconnected regions begin in nearly the same state?

This is the horizon problem. The causal scale at recombination corresponds to only a small part of the observed sky.

Inflation stretches one causal patch to cosmic size

Inflation is a period of accelerated expansion,

$$\ddot{a} > 0 \quad \Longleftrightarrow \quad \frac{d}{dt} \left(\frac{1}{aH} \right) < 0.$$



The hot thermal evolution begins inside this enlarged, nearly homogeneous region.

The regions we see had a common causal past before inflation stretched them apart.

$H \equiv \dot{a}/a$; the essential feature is a shrinking comoving Hubble radius.

Inflation also provides the primordial perturbations

The field driving inflation cannot be perfectly smooth: quantum fluctuations are stretched to macroscopic wavelengths.

$$\delta\phi_k \sim \frac{H}{2\pi} \quad \longrightarrow \quad \mathcal{R}(\mathbf{k}) \simeq -\frac{H}{\dot{\phi}} \delta\phi_{\mathbf{k}}.$$

- $\mathcal{R}(\mathbf{k})$ is the primordial curvature perturbation for spatial wavevector $\mathbf{k}$.
- It is approximately conserved while the wavelength is larger than the Hubble radius.
- After re-entry, it seeds gravitational potential wells and acoustic motion.

Inflation turns microscopic fluctuations into the initial conditions for cosmic structure.

Single-field slow-roll illustration. The exact amplitude and evolution depend on the inflationary model.

A simple inflationary picture makes testable predictions

$$\Delta_{\mathcal{R}}^2(k) = A_s \left(\frac{k}{k_*} \right)^{n_s-1}.$$

Initial statistics

Nearly scale invariant, nearly Gaussian, and predominantly adiabatic.

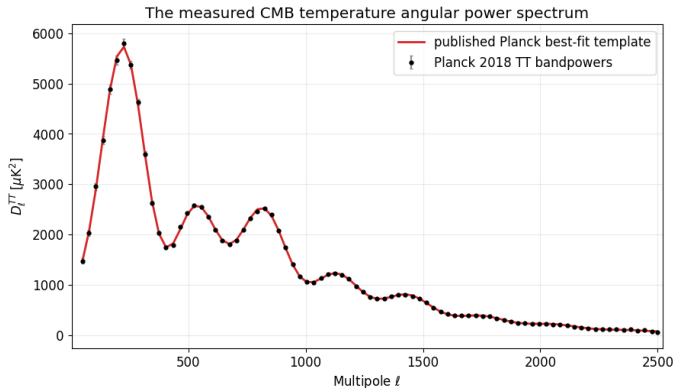
Coherent evolution

Each Fourier mode begins in the same growing mode, so its acoustic phase is synchronized.

Inflation may also produce tensor perturbations, described by the tensor-to-scalar ratio r .

Coherent inflationary initial conditions make the acoustic peak sequence possible.

This is the pattern we want to explain



How much temperature variation is present at each angular scale? Larger multipoles correspond to smaller angles. Real Planck 2018 TT bandpowers and published best-fit template, from the Lecture 2 notebook; formal definitions come in Lecture 2.

A nearly uniform blackbody sky

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/(k_B T)} - 1}, \quad T_0 \simeq 2.725 \text{ K.}$$

$$T(\hat{n}) = T_0[1 + \Theta(\hat{n})], \quad \Theta = \frac{\delta T}{T_0}.$$

After removing the motion-induced dipole, typical temperature variations are of order 10^{-5} .

The mean temperature records thermal history. The small variations record gravity and the contents of the Universe.

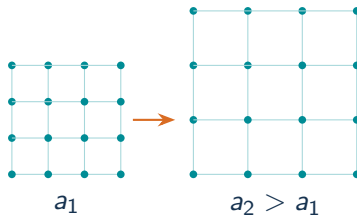
Convention: $c = \hbar = k_B = 1$ unless units are explicitly restored.

Expansion changes physical distances

$$ds^2 = -dt^2 + a^2(t)dx^2$$

$$d_{\text{phys}} = a(t)|\Delta \mathbf{x}|$$

Comoving coordinates $\mathbf{x}$ label the same freely moving observers. We choose $a(t_0) = 1$.



What controls the expansion?

$$H \equiv \frac{1}{a} \frac{da}{dt}, \quad H^2 = \frac{8\pi G}{3} \rho$$

$$\frac{1}{a} \frac{d^2 a}{dt^2} = -\frac{4\pi G}{3} (\rho + 3P).$$

- Density determines the expansion rate.
- Pressure also gravitates.
- H^{-1} gives the instantaneous expansion timescale; the age follows from the full expansion history.

Spatially flat FLRW background; t is cosmic time.

Local conservation does not mean constant total energy

$$\nabla_\mu T^{\mu\nu} = 0 \quad \Longrightarrow \quad \frac{d\rho}{dt} + 3H(\rho + P) = 0.$$

For a fixed comoving region, $V \propto a^3$ and $E = \rho V$:

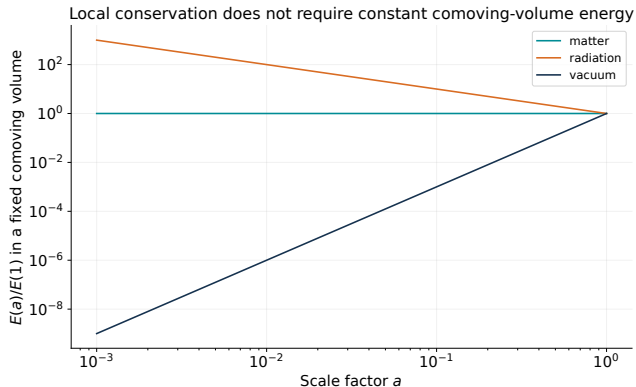
$$dE = -P dV.$$

A general expanding FLRW spacetime lacks the time-translation symmetry that would define a globally conserved energy of the usual kind.

For a separately conserved fluid with $P = w\rho$,

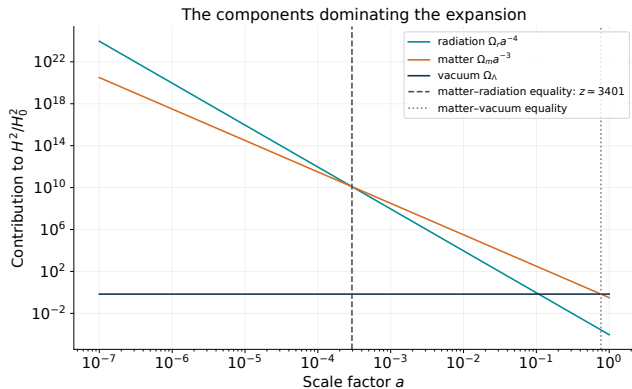
$$\rho \propto a^{-3(1+w)}, \quad E \propto a^{-3w}.$$

Energy inside the same comoving region



Each component is normalized to its own present energy. The curves compare how the components evolve; their relative heights do not show their abundances.

Different components dominate at different times



Lecture 1 notebook. Matter-radiation equality is where $\Omega_r a^{-4} = \Omega_m a^{-3}$.

Expansion cools the photon bath

$$\lambda \propto a, \quad E_\gamma \propto a^{-1}, \quad 1 + z = \frac{1}{a}.$$

Uniform redshifting preserves a Planck spectrum:

$$T(z) = T_0(1 + z).$$

At $z \simeq 1100$, $T \simeq 3000$ K, or $k_B T \simeq 0.26$ eV.

Define distances

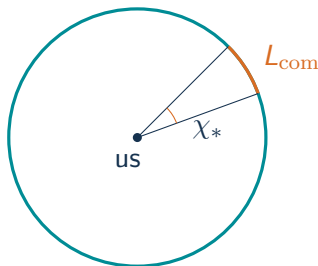
$$d\tau = \frac{dt}{a}, \quad \chi = |\mathbf{x}|$$

For incoming radial light: $d\chi/d\tau = -1$.

$$\chi_* = \tau_0 - \tau_* = \int_0^{z_*} \frac{dz}{H(z)}.$$

L_{com} : transverse comoving size of a feature on the last-scattering shell.

$$\theta \simeq \frac{L_{\text{com}}}{\chi_*} \quad (\text{small angle})$$



last-scattering shell

Later, our physical ruler will be the sound horizon.

From now on, dots denote $d/d\tau$ and $\mathcal{H} = \dot{a}/a = aH$. Distances above use $c = 1$.

There is more than one useful “horizon”

$$H \equiv \frac{1}{a} \frac{da}{dt}, \quad \boxed{\mathcal{H} \equiv \frac{1}{a} \frac{da}{d\tau} = aH.}$$

H uses cosmic time t ; $\mathcal{H}$ uses conformal time τ .

Comoving Hubble length $\mathcal{H}^{-1}$: the expansion timescale expressed as a comoving length.

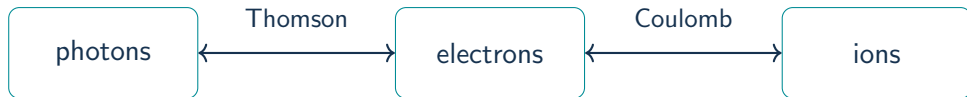
Comoving particle horizon $\int_{t_i}^t dt'/a(t')$: how far light could have travelled.

Sound horizon $r_s(\tau) = \int_0^\tau c_s(\tau') d\tau'$: how far a pressure disturbance could have travelled.

For acoustic oscillations, the relevant causal scale is the sound horizon.

Why was the early Universe opaque?

$$\Gamma_T = n_e \sigma_T, \quad \lambda_{\text{mfp}} = (n_e \sigma_T)^{-1}.$$



$\Gamma \gg H$: frequent interactions; $\Gamma \lesssim H$: decoupling.

How does equilibrium divide hydrogen between ions and atoms?

$$p + e^- \leftrightarrow H^0 + \gamma, \quad \mu_p + \mu_e = \mu_{H^0}, \quad \mu_\gamma = 0.$$

For a dilute non-relativistic species, including rest energy in μ_i ,

$$n_i = g_i \left(\frac{m_i T}{2\pi} \right)^{3/2} e^{(\mu_i - m_i)/T}.$$

- n_e and n_p count *free* electrons and protons.
- n_{H^0} counts neutral hydrogen atoms, a separate species.
- g_i counts the internal states of species i .

Form $n_e n_p / n_{H^0}$: chemical equilibrium will eliminate the chemical potentials.

Hydrogen only; natural units. The full derivation is in Section 5.2 of the notes.

Use equilibrium to relate the three species

Insert the three species densities:

$$\frac{n_e n_p}{n_{H^0}} = \frac{g_e g_p}{g_{H^0}} \left(\frac{m_e m_p}{m_{H^0}} \frac{T}{2\pi} \right)^{3/2} e^{(\mu_e + \mu_p - \mu_{H^0})/T} e^{-(m_e + m_p - m_{H^0})/T}.$$

- Chemical equilibrium: $\mu_e + \mu_p - \mu_{H^0} = 0$.
- Binding energy: $m_e + m_p - m_{H^0} = E_I = 13.6 \text{ eV}$.
- In the prefactor only, $m_{H^0} \simeq m_p$; ground-state spin counts give $g_e g_p / g_{H^0} = 2 \times 2 / 4 = 1$.

$$\frac{n_e n_p}{n_{H^0}} \simeq \left(\frac{m_e T}{2\pi} \right)^{3/2} e^{-E_I/T}.$$

The tiny binding-energy mass difference must be retained in the exponential, where it supplies the temperature dependence.

Count every hydrogen nucleus

$$n_H = \underbrace{n_p}_{\text{free protons}} + \underbrace{n_{H^0}}_{\text{neutral atoms}}$$

Each neutral atom contains one proton: n_H counts *all* hydrogen nuclei.

Define the ionized fraction using this total; charge neutrality gives $n_e = n_p$:

$$x_e = \frac{n_e}{n_H}, \quad n_e = n_p = x_e n_H, \quad n_{H^0} = (1 - x_e) n_H.$$

Substitute into the ratio:

$$\frac{n_e n_p}{n_{H^0}} = \frac{(x_e n_H)^2}{(1 - x_e) n_H} = n_H \frac{x_e^2}{1 - x_e}.$$

Divide the preceding Saha ratio by n_H to obtain an equation for x_e .

Only in the fully ionized limit is $n_H \simeq n_p$. The ratio $n_e/n_p = 1$ follows from charge neutrality and therefore does not measure ionization.

Why recombination occurs far below 13.6 eV

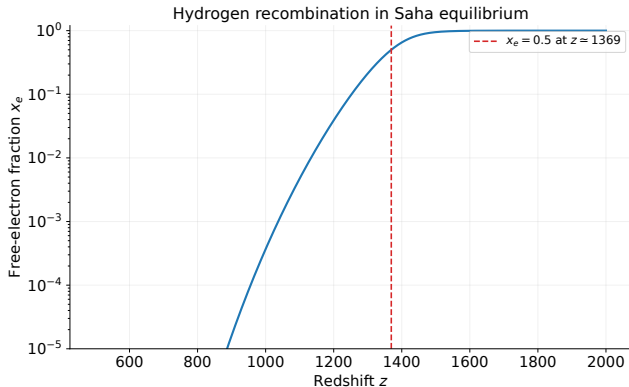
Combining the equilibrium ratio with the ionized fraction gives the *Saha equation*:

$$\frac{x_e^2}{1 - x_e} = \frac{1}{n_H} \left(\frac{m_e T}{2\pi} \right)^{3/2} e^{-E_I/T}.$$

- At $T = E_I$, the density-dependent prefactor is enormous: $x_e \simeq 1$.
- Roughly a billion photons per baryon keep the ionizing tail populated.
- Half ionization requires $E_I/T \simeq 42$, giving $T \simeq 0.32$ eV.

This equilibrium result determines the ionization fraction. Last scattering also depends on the interaction rate and expansion.

Neutral hydrogen forms rapidly as the plasma cools



Hydrogen Saha equilibrium. The cosmological evolution proceeds from right to left on this redshift axis.

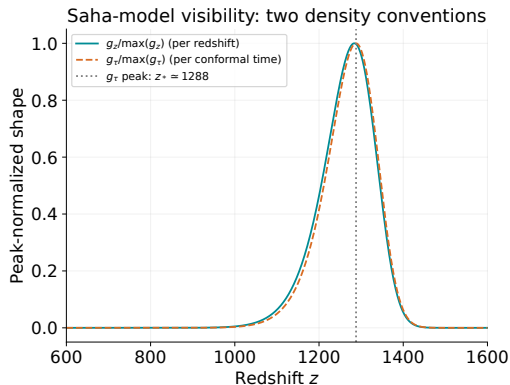
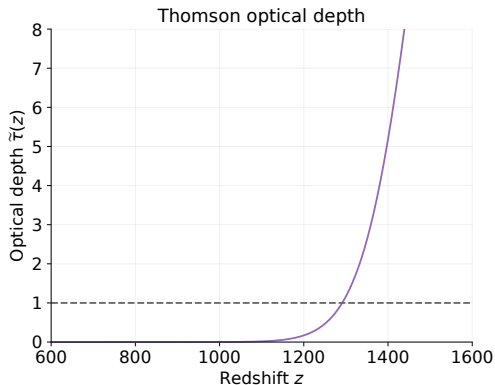
Last scattering is a probability distribution

$$\tilde{\tau}(\tau) = \int_{\tau}^{\tau_0} a n_e \sigma_T d\tau', \quad \dot{\tilde{\tau}} < 0.$$

- $e^{-\tilde{\tau}}$ is the probability of reaching us without another scattering.
- The visibility is $\boxed{g_{\tau} = -\dot{\tilde{\tau}} e^{-\tilde{\tau}}}$.
- The finite visibility width corresponds to a last-scattering shell with finite thickness.

The tilde matters: $\tilde{\tau}$ is optical depth; τ is conformal time.

Recombination and last scattering are related, but distinct



Saha-model demonstration. $g_z = g_\tau/H(z)$, so the peak depends on the integration variable. Precision calculations place last scattering near $z_* \simeq 1100$; this equilibrium toy model peaks earlier.

Ten-minute break

We have an expanding, cooling Universe that becomes transparent.
What could make different patches of the last-scattering shell have different temperatures?

Add small perturbations to the smooth background

$$\rho_i(\tau, \mathbf{x}) = \bar{\rho}_i(\tau)[1 + \delta_i(\tau, \mathbf{x})].$$

In conformal Newtonian gauge,

$$ds^2 = a^2 \left[- (1 + 2\Psi) d\tau^2 + (1 - 2\Phi) d\mathbf{x}^2 \right].$$

- Ψ : gravitational potential; a potential well has $\Psi < 0$.
- Φ : scalar perturbation of spatial curvature.
- Keep terms linear in potentials, density contrasts, and velocities.

Intermezzo: Why do two scalar potentials remain?

$$\begin{aligned} & 4 \text{ scalar metric functions} \\ & - 2 \text{ scalar coordinate freedoms} \\ & = 2 \text{ scalar metric potentials} \end{aligned}$$

These are constrained scalar potentials. The two propagating gravitational-wave polarizations belong to the tensor sector.

Einstein's equations relate the scalar potentials to matter perturbations.

Anisotropic stress is directional momentum flux

$$T^i_j = P\delta^i_j + \pi^i_j, \quad \pi^i_i = 0.$$

- A quadrupole of particle momenta gives stress beyond isotropic pressure.
- Free-streaming neutrinos arrive from different regions without frequent collisions to isotropize them.
- Tight coupling suppresses photon anisotropic stress; photons develop it as scattering weakens.

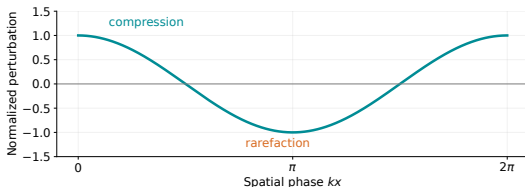
Negligible total scalar anisotropic stress gives $\Phi = \Psi$. Cold matter has small random-velocity stresses.

Why switch from position $\mathbf{x}$ to wave number $\mathbf{k}$?

The perturbation contains many spatial scales. Because the equations are linear, we may evolve one sinusoidal component at a time:

$$\delta(\mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \delta(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{x}}.$$

- $\mathbf{k}$ gives the direction; $k = |\mathbf{k}|$ gives the scale.
- The wavelength is $\lambda = 2\pi/k$: large k means a short wave.
- Derivatives become algebraic: $\nabla \rightarrow i\mathbf{k}$ and $\nabla^2 \rightarrow -k^2$.



The drawing is one mode at one instant: crests are compressions and troughs are rarefactions. We will solve for how its amplitude changes with time.

Plan for deriving the acoustic wave

For one Fourier mode, we track

$$\underbrace{\Theta_0 = \delta_\gamma/4}_{\text{photon monopole: density}}, \quad \underbrace{\theta_i = i\mathbf{k} \cdot \mathbf{v}_i}_{\text{velocity divergence}} \quad (i = \gamma, b), \quad \underbrace{\Phi, \Psi}_{\text{gravity}}.$$

For photons, θ_γ carries the same bulk-flow information as the temperature dipole Θ_1 .

Two evolution equations

Continuity: motion changes density.

Euler: pressure and gravity change motion.

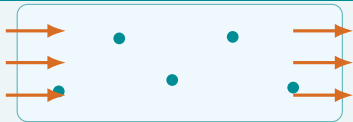
Scattering and baryons

Scattering drives θ_γ toward θ_b . Baryons then add inertia to the pressure-supported photon fluid.

We will combine continuity and Euler into one oscillator equation. Pressure restores, gravity drives, and baryons change the inertia.

Velocity moves the fluid; convergence compresses it

Uniform translation

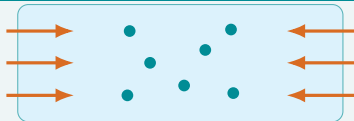


The same amount enters and leaves:

$$\mathbf{v}_\gamma \neq 0, \text{ but } \nabla \cdot \mathbf{v}_\gamma = 0.$$

The patch moves, but its density is unchanged.

Converging flow



More enters than leaves:

$$\theta_\gamma \equiv \nabla \cdot \mathbf{v}_\gamma < 0.$$

Photons accumulate, so the patch becomes denser and hotter.

Continuity counts net flow through a boundary: $\dot{\Theta}_0 = -\theta_\gamma/3 + \dot{\Phi}$. The velocity matters through its spatial variation.

Baryon loading: photon pressure moves two fluids

Momentum density is proportional to $(\rho + P)\mathbf{v}$.

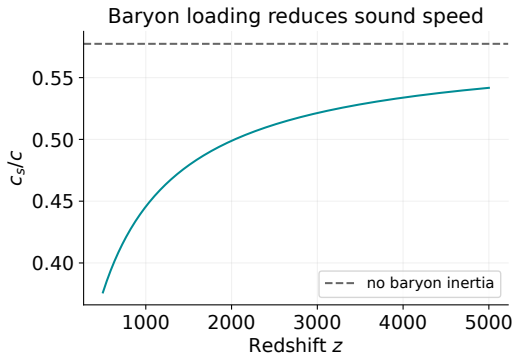
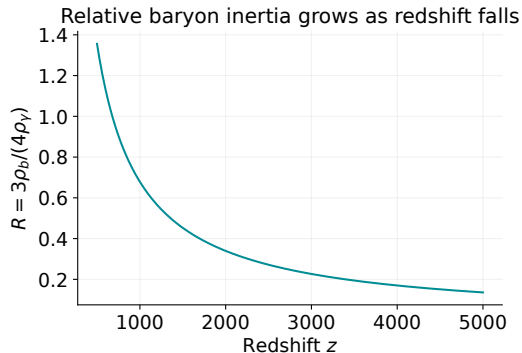
$$\boxed{R = \frac{\bar{\rho}_b}{4\bar{\rho}_\gamma/3}} \implies \text{total inertia} = \frac{4\bar{\rho}_\gamma}{3}(1 + R).$$

Photons supply the pressure while photons and baryons both have to accelerate. For an adiabatic compression, $\delta\rho_b = R\delta\rho_\gamma$, hence

$$c_s^2 = \frac{\delta P_\gamma}{\delta\rho_\gamma + \delta\rho_b} = \boxed{\frac{1}{3(1 + R)}}.$$

R is the extra baryon inertia measured in units of photon inertia. Larger R lowers the sound speed.

As the Universe expands, baryons become harder to move



The left panel shows $R = 3\bar{\rho}_b/(4\bar{\rho}_\gamma)$; the right shows $c_s/c = [3(1 + R)]^{-1/2}$. Cosmic time runs from high redshift on the right toward low redshift on the left.

Step 1: converging motion compresses the photons

$$\dot{\Theta}_0 = -\frac{1}{3}\theta_\gamma + \dot{\Phi}.$$

- $\Theta_0 = \delta_\gamma/4$ measures the photon-temperature monopole.
- $\theta_\gamma = i\mathbf{k} \cdot \mathbf{v}_\gamma < 0$ means that the photon flow converges, so Θ_0 rises.
- $\dot{\Phi}$ accounts for compression caused by changing spatial curvature.

The equivalent density equation is $\dot{\delta}_\gamma = -4\theta_\gamma/3 + 4\dot{\Phi}$.

Step 2: pressure and gravity set photons in motion

$$\dot{\theta}_\gamma = \underbrace{k^2 \Theta_0}_{\text{pressure}} + \underbrace{k^2 \Psi}_{\text{gravity}} + \underbrace{\Gamma_c (\theta_b - \theta_\gamma)}_{\text{scattering}},$$

$$\Gamma_c \equiv -\dot{\tilde{\tau}} = a n_e \sigma_T > 0.$$

- A temperature crest has higher photon pressure, which pushes material away from it.
- Gravity pulls the fluid into potential wells.
- Scattering acts whenever the photon and baryon flows differ.

There is no photon Hubble drag: $\mathcal{H}(1 - 3w_\gamma)\theta_\gamma = 0$ because $w_\gamma = 1/3$. The signs look different from vector forces because θ is a divergence. Leading tight coupling neglects the photon quadrupole; $\Gamma_c = a\Gamma_T$.

Step 3: scattering transfers momentum between the fluids

$$\dot{\theta}_b + \underbrace{\mathcal{H}\theta_b}_{\text{expansion drag}} = k^2\Psi + \frac{\Gamma_c}{R}(\theta_\gamma - \theta_b).$$

Baryon pressure $c_b^2 k^2 \delta_b$ is negligible because $c_b^2 \sim T_b/m_b \ll 1$; photon pressure is transmitted by scattering.

The photon–baryon *slip* measures their relative flow:

$$\Delta\theta \equiv \theta_\gamma - \theta_b.$$

The collision terms exchange momentum internally:

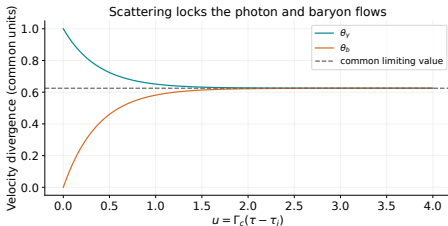
$$\dot{\theta}_\gamma \Big|_{\text{coll}} + R \dot{\theta}_b \Big|_{\text{coll}} = 0.$$

Scattering reduces the slip. The same force density gives a baryon acceleration proportional to $1/R$ because R measures baryon inertia.

Tight coupling means that the slip rapidly decays

Keeping only the collision terms gives

$$\frac{d\Delta\theta}{d\tau} = -\Gamma_c(1 + R^{-1})\Delta\theta.$$



Here $u = \Gamma_c(\tau - \tau_i)$ counts elapsed scattering times in a fixed- R , fixed-rate collision-only example. The two flows approach their inertia-weighted common value.

After this short transient, $\theta_\gamma \simeq \theta_b \equiv \theta$. We can then describe one tightly coupled fluid.

Step 4: one momentum equation for the coupled plasma

Start from the two equations, weighting the baryon equation by its inertia:

$$\text{photon Euler: } \dot{\theta}_\gamma = k^2 \Theta_0 + k^2 \Psi + \Gamma_c(\theta_b - \theta_\gamma),$$

$$R \times \text{baryon Euler: } R\dot{\theta}_b + \mathcal{H}R\theta_b = Rk^2 \Psi + \Gamma_c(\theta_\gamma - \theta_b).$$

Add them: the collision terms cancel. With $\theta_\gamma \simeq \theta_b \equiv \theta$,

$$(1 + R)\dot{\theta} + \mathcal{H}R\theta = k^2 \Theta_0 + (1 + R)k^2 \Psi.$$

Using $\dot{R} = \mathcal{H}R$ and dividing by $1 + R$ gives

$$\dot{\theta} + \frac{\dot{R}}{1 + R}\theta = \frac{k^2}{1 + R}\Theta_0 + k^2 \Psi.$$

Photon pressure is divided by the total inertia $1 + R$, while gravity accelerates both components.

The slip is small, but $\Gamma_c(\theta_\gamma - \theta_b)$ remains the finite internal force that transfers photon momentum to baryons.

Step 5: continuity and Euler give an acoustic oscillator

Differentiate continuity and use the combined momentum equation to eliminate θ :

$$\ddot{\Theta}_0 + \underbrace{\frac{\dot{R}}{1+R}\dot{\Theta}_0}_{\text{changing inertia}} + \underbrace{k^2 c_s^2 \Theta_0}_{\text{pressure restoring force}} = \underbrace{S_{\text{grav}}}_{\text{gravitational driving}}$$

$$S_{\text{grav}} = \ddot{\Phi} + \frac{\dot{R}}{1+R}\dot{\Phi} - \frac{k^2}{3}\Psi.$$

- The natural frequency is approximately kc_s : short waves oscillate faster.
- Baryon inertia lowers c_s and changes the amplitude.
- Constant or evolving potentials displace or drive the oscillation.

Why introduce an effective temperature?

The emitting plasma has intrinsic perturbation Θ_0 .

A photon leaving a static potential acquires fractional shift $\Psi_* - \Psi_{\text{obs}}$.

The observer term is common to all directions. The emission contribution is

$$X = \Theta_0 + \Psi.$$

X is the temperature perturbation at emission after including the gravitational redshift from the emitting potential.

Doppler shifts and changing potentials along the path will enter Lecture 2.

Make the driving and equilibrium explicit

$$\ddot{X} + \frac{\dot{R}}{1+R}\dot{X} + k^2 c_s^2 X = \ddot{\Phi} + \ddot{\Psi} + \frac{\dot{R}}{1+R}(\dot{\Phi} + \dot{\Psi}) - k^2 c_s^2 R\Psi.$$

Freeze R, Φ, Ψ for a simple solvable model:

$$\ddot{X} + k^2 c_s^2 (X + R\Psi) = 0.$$

For fixed R and Ψ , the acceleration vanishes at $X_{\text{eq}} = -R\Psi$. Thus X oscillates around $-R\Psi$, rather than around zero as it would without baryons.

Freezing coefficients is a toy approximation. Expansion-related amplitude evolution is distinct from dissipative diffusion damping.

Derive the equilibrium from force balance

No velocity or acceleration means pressure balances gravity:

$$0 = \frac{k^2}{1+R} \Theta_{0,\text{eq}} + k^2 \Psi.$$

$$\Theta_{0,\text{eq}} = -(1+R)\Psi,$$

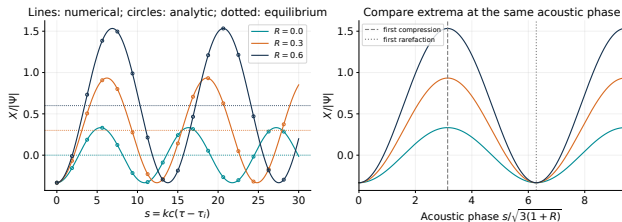
$$\boxed{X_{\text{eq}} = -R\Psi.}$$

- Photons must support their own inertia *and* the baryons.
- For $R = 0$, intrinsic temperature and gravitational redshift cancel at equilibrium.
- For $\Psi < 0$ and $R > 0$, the effective-temperature offset is positive.

How to read the baryon-loading toy model

The vertical axis divides by $|\Psi|$ to remove the arbitrary amplitude of this linear mode.

$$s = k(\tau - \tau_i), \quad \varphi = kr_s = \frac{s}{\sqrt{3(1+R)}}.$$



- **Left, same s :** larger R oscillates more slowly.
- **Right, same φ :** aligned cycles expose the shifted equilibrium and unequal extrema.

s measures elapsed time in wavelength units; $\varphi = kr_s$ is the acoustic phase derived from the sound speed. Lines solve one Fourier mode; circles show the analytic solution.

The sound horizon keeps track of acoustic phase

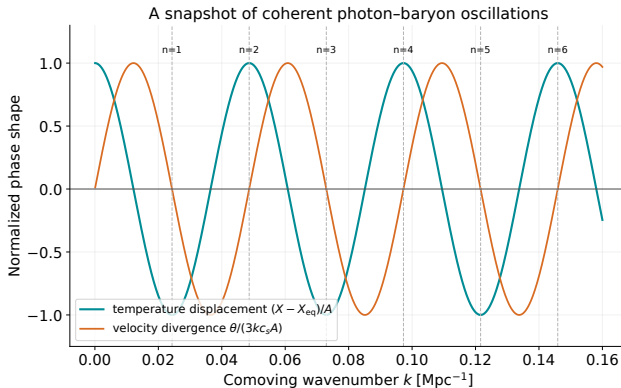
$$r_s(\tau) = \int_0^\tau c_s d\tau', \quad X + R\Psi = C_1 \cos(kr_s) + C_2 \sin(kr_s).$$

- A displaced fluid with negligible initial velocity starts in the cosine phase.
- Adiabatic growing modes have coherent temporal phases.
- Recombination takes a snapshot at $r_s(\tau_*)$.

Extrema satisfy $k_n r_s(\tau_*) \simeq n\pi$: a harmonic series.

The expression is exact for frozen coefficients; slowly evolving coefficients also change the amplitude and equilibrium.

Temperature and velocity are out of phase



The horizontal coordinate is acoustic phase kr_s . The curves show normalized phase relations; their vertical scales do not represent equal observable amplitudes.

Projection and diffusion set the final angular pattern

$$\theta_s = \frac{r_s(\tau_*)}{\chi_*}, \quad \ell_A \simeq \frac{\pi}{\theta_s}.$$

- Sound supplies a ruler; the distance turns it into an angle.
- Baryon loading modulates compression versus rarefaction peaks.
- Potentials decaying near equality boost early-entering modes: radiation driving.
- Photon diffusion damps short waves: amplitude $\sim e^{-k^2/k_D^2}$.
- Finite last-scattering width averages over nearby phases.

ℓ_A gives the characteristic spacing. Potential evolution and projection shift the individual peaks, including the first.

Explore the physics in the notebook

Open Lecture 1 in Google Colab

1. Compare 13.6 eV with the Saha half-ionization temperature.
2. Vary R and compare the final common velocity.
3. Increase baryon loading: what happens to frequency and equilibrium?

Change one input at a time and connect the result to the equations.

All plots are included in these slides, so the session also works without an internet connection.

The physical story in five steps

1. Expansion cools the photon bath.
2. Neutral-atom formation reduces opacity.
3. Gravity and photon pressure create acoustic oscillations.
4. Baryons shift equilibrium; diffusion damps short scales.
5. Last scattering projects a snapshot onto our sky.

Next: derive the angular power spectrum and measure it with real data.